Recognising Spoken Numbers from Synthetic Base Words Only: an Exploratory Study with Frozen Speech Encoders

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Abstract

Can a recogniser learn all spoken numbers from 0 to 99 when it has only heard the base words of the numeration, spoken by synthetic voices? We train a small CTC head on a frozen pre-trained speech encoder with the 23 French base words, produced by a text-to-speech (TTS) engine, together with random sequences of these words that are sped up and cross-faded; every sequence that contains a valid compound number is removed from task-specific training and validation. The frozen encoder was previously trained on speech; the restriction does not apply to its pre-training. A grammar restricted to the 100 valid word sequences decodes the output. On 928 real recordings from 45 speakers covering numbers 0–89, a frozen Whisper encoder with a transformer head recognises 86.7 % of the numbers (95 % interval over speakers: 82.8–91.0 %) and 77.3 % of the compound numbers, against 31.3 % for the same model trained on isolated words only and 95.5 % when full synthetic numbers are available. The remaining errors concentrate on liaison forms (*vingt-et-un*, 18.6 %) and on *quatre-vingt* (44.6 %). An exploratory transfer to English reaches 96.0 % on real speakers, whose compound numbers come from three speakers only. Code, trained heads and the real-recording list will be released at a later date. The real evaluation set was consulted during method development; these are exploratory results, not a confirmatory held-out benchmark.

1 Introduction

Speech recognition systems generally require large amounts of transcribed audio data for training. For limited-vocabulary tasks, such as recognising numbers or voice commands, collecting recordings from many speakers can be costly. Speech synthesis now makes it possible to generate different voices, while pre-trained speech encoders, such as Whisper and wav2vec 2.0, can be adapted to specific tasks [1, 2]. Reducing data requirements and computational costs has also become an important challenge [24, 25, 23].

In French, numbers from 0 to 99 can be constructed using only 23 basic words. For example, *quatre-vingt-dix-sept* (97) combines four words. Recognising numbers from isolated words has already been studied [16]. In this work, we investigate whether a model can recognise compound numbers without using them during task-specific training. To do so, we use synthetic voices, filtered artificial word sequences, and frozen pre-trained speech encoders.

Our main contributions are:

- A method (Fig. 1) achieving 77.3 % accuracy on real spoken compound numbers, compared with 31.3 % when training on isolated words only, without using compound numbers during training or validation.
- A fully synthetic training protocol, with separate voices for training, validation, and synthetic testing, also evaluated on 928 real recordings from 45 speakers. Unlike existing datasets [12, 14, 4], we evaluate a model trained exclusively on synthetic data using real human speech.
- A comparison with models trained on complete numbers, along with an analysis of errors, noise robustness, data and compute requirements, and exploratory transfer to English and Arabic.

2 Related work

Synthetic data. TTS augmentation improves ASR when mixed with real data [5, 6], including for numeric sequences [7], while training on synthetic speech alone leaves a gap [8]; keyword spotters built on a pre-trained speech embedding can be trained on synthesized speech alone [9]. **Pre-trained encoders.** Self-supervised wav2vec 2.0 [1] (here the French LeBenchmark model [10]) gives representations that transfer to many tasks [3], as does weakly supervised Whisper [2]; phonetic and word information peaks in intermediate layers [11]. **Small vocabularies and composition.** Digit and keyword datasets (TIDIGITS [12], Speech Commands [13], AudioMNIST [14], MSWC [15]) treat each item as a class; SpokeN-100 [4] provides 0–99 in four languages, including French, with synthetic voices. Connected digits were recognised from isolated-word templates by level-building DTW [16]. Neural models often fail to generalise systematically to unseen combinations [17]. CTC [18] with a grammar or WFST constraint is standard in ASR [19].

3 Method

Words and grammar. Each number n maps to a word sequence $g(n)$ over a vocabulary V (French: 0–16, the tens *vingt*…*soixante* and *et*; $|V| = 23$); the map is injective on 0–99. 22 numbers are a single word; the 78 others are compounds. **Training data (words only).** Each word of V is synthesised in isolation by 8 TTS voices with 4 prosody variants (736 clips). We draw 3,000 random sequences of 2–4 words of one voice and variant, in arbitrary order (e.g. *sept trente douze*); each word is time-stretched by a factor drawn in 1.1–1.9 and cross-faded (20–60 ms) with the next. Any sequence that contains a valid compound as a contiguous

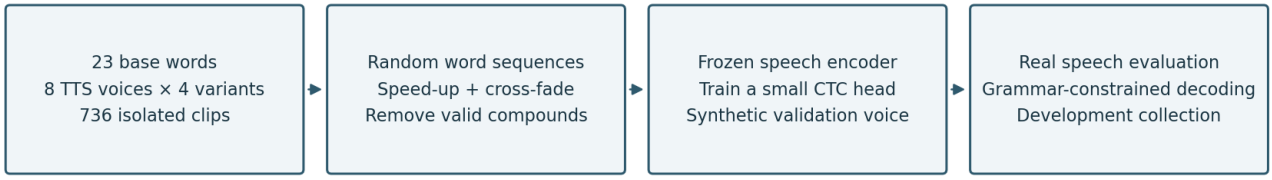


Figure 1: Recipe. Only the base words are synthesised; random sequences of sped-up, cross-faded words, in arbitrary order and with every sequence containing a valid compound removed, train a small CTC head on a frozen encoder. The grammar is used both to filter training/validation sequences and to constrain decoding. The acoustic head receives no compound training example.

sub-sequence (e.g. *sept vingt trois* contains 23) is removed: 533 sequences (17.8 %), leaving 2,467. Training therefore contains no compound number. **Model.** A frozen encoder gives frame features: the Whisper-base encoder [2] (last layer), wav2vec2-FR-7K layer 8 [10], or a log-mel front-end with a 3-block CNN. A head outputs per-frame posteriors over V plus the CTC blank: a 2-layer transformer (4 heads, d=128; 0.47 M parameters on Whisper, 0.50 M on wav2vec2) or a 2-layer BiGRU. The words-only Whisper transformer is trained for 40 epochs with cross-fades and 80 epochs with isolated words only. Other words-only heads use 40 epochs; full-number heads use 30 epochs for mel and 25 for frozen encoders. Padded frames are masked (attention mask, packed sequences), so a prediction does not depend on the other inputs of a batch. For frozen encoders, only the head is trained; the mel baseline also trains its CNN. Training uses the CTC loss and SpecAugment-style masking [20]. **Decoding.** We keep the valid number whose word sequence is most likely under the CTC model,

$$\hat{n} = \arg\max_{n \in \{0..99\}} \log p_{\mathrm{CTC}}\big(g(n) \mid h_{1:T}\big). \qquad (1)$$

Model selection uses word sequences of a held-out voice (fr_henri), filtered in the same way: 246 of 300 generated validation sequences remain. AdamW uses a maximum learning rate of 10^{-3}, weight decay 10^{-3}, a OneCycle schedule, batches of 32 and gradient clipping at 5. The epoch with highest exact greedy validation-sequence accuracy is retained; ties select the later epoch.

4 Experimental setup

Synthetic data. edge-tts neural voices: 13 French voices (France, Belgium, Switzerland, Canada), 8 for training, 1 for validation, 4 unseen test voices (one per country). Full numbers synthesised as single utterances (4 prosody variants) are used to test on unseen voices and, as an upper reference, to train. **Real test.** Lingua Libre [21] volunteer recordings (CC BY-SA), filtered (non-audio files, alternative pronunciations, mismatched titles, speakers with fewer than 3 numbers). French: 928 recordings, 45 speakers, numbers 0–89; 475 are single-word numbers and 453 compounds, the latter from 13 speakers. The file list is fixed and checked at every run. No real recording enters gradient updates or epoch selection. However, real scores informed method development, so the collection serves as a development evaluation set. **Statistics.** We report the mean ± standard deviation over training seeds (5 for the words-only Whisper transformer and all English and Arabic runs; 3 for other words-only configurations and every full-number configuration), a 95 % interval obtained by resampling speakers (2,000 bootstrap draws), and, where useful, the accuracy averaged per number so that frequent numbers do not dominate.

Table 1: French, trained on the base words only (no compound number heard). Accuracy (%) on the 453 real compound recordings and on all 928 real recordings; synthetic = compounds by 4 unseen TTS voices.

Data	Encoder, head (seeds)	Synthetic	Real comp.	Real all
isolated words	Whisper, transformer (5)	42.7±8.3	31.3±5.1	64.7±2.6
+ cross-faded seq.	mel, BiGRU (3)	28.5±1.5	13.3±2.7	48.8±0.8
+ cross-faded seq.	wav2vec2, BiGRU (3)	28.5±14.0	21.0±8.0	55.2±5.1
+ cross-faded seq.	Whisper, BiGRU (3)	67.8±31.6	54.1±26.8	74.9±14.8
+ cross-faded seq.	wav2vec2, transformer (3)	91.1±3.2	68.7±1.3	81.8±1.1
+ cross-faded seq.	Whisper, transformer (5)	**94.7±0.9**	**77.3±1.8**	**86.7±1.1**

5 Results

5.1 Words only

Table 1 gives the ablation. With isolated words only, the Whisper model recognises single words well (96.5 %) but only 31.3 % of compounds. With cross-faded sequences, compound accuracy is 77.3 % (95 % interval over speakers: 71.9–82.1 %) and all numbers to 86.7 % (82.8–91.0 %); per number, 81.3 %. Both the head and the encoder matter. BiGRU heads fit the artificial sequences (92–98 % exact sequences on the held-out voice) but transfer poorly to natural numbers (13–54 % on real compounds), and on Whisper the BiGRU result depends strongly on the seed (23, 66 and 73 %). With the transformer head, wav2vec2 reaches 68.7 % and Whisper 77.3 %, both stable across seeds. The grammar matters: greedy decoding gives 66.1 % on real compounds.

5.2 Reference: training on full synthetic numbers

With full numbers (Table 2), the best system reaches 95.5 % (92.8–97.7 % over speakers). With the same encoder and

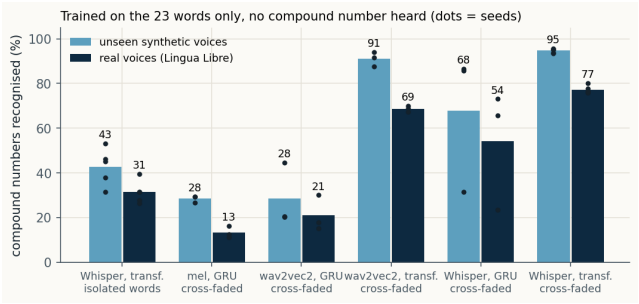


Figure 2: French compound numbers recognised after training on the base words only.

Table 2: French, trained on the 3,200 full synthetic numbers of the 8 training voices (upper reference), 3 seeds. Real = 928 recordings; chance = 1 %.

Model	TTS unseen	Real	Real, greedy	Real, per number
mel + 100-class	74.0±1.8	47.8±1.6	–	41.1
mel + CRNN-CTC	90.8±1.2	71.9±0.4	68.3	69.7
wav2vec2 + CTC (GRU)	99.5±0.2	90.5±0.9	88.8	89.5
wav2vec2 + 100-class	99.3±0.2	92.0±0.3	–	91.8
Whisper + CTC (GRU)	99.9±0.1	94.4±0.3	93.0	92.0
Whisper + CTC (transformer)	99.8±0.2	**95.5±0.4**	92.2	**94.9**
Whisper-base zero-shot	54.4	34.4	–	–

head, training on words only costs 8.9 points on all numbers and 17.9 points on compounds (95.1 vs 77.3 %). The synthetic-to-real gap is small for frozen pre-trained encoders (4–9 points) and large for the mel model (19–26 points). Off-the-shelf Whisper-base transcription fails on isolated short numbers, while its encoder is the best frozen feature extractor here. **Unseen numbers.** With 10 numbers removed from training and validation (23, 31, 38, 45, 57, 64, 76, 83, 92, 99), word-level CTC models still recognise them on unseen synthetic voices (mel 86.7±3.4 %, wav2vec2 99.2±1.0 %, 3 seeds), whereas the 100-class models, whose 10 corresponding outputs received no example, recognise none of them (0 % for mel and wav2vec2, measured).

5.3 Error analysis

Errors concentrate on two groups (Table 3). The most frequent confusions over the 5 seeds drop the final *et un*: 21→20 (41 times), 51→50 (30), 31→30 (25), 41→40 (23); others drop the first word: 19→9 (16), 27→7 (12), 84→4 (11); 61→60 (15) also drops *et un*.

5.4 Robustness to noise

With real background noise (Table 4), the Whisper-based models lose 1–7 points at 20–10 dB; at 0 dB the words-only model (70.6 %) is close to the full-number model (71.2 %). The mel model collapses. All models degrade strongly when competing voices say numbers at 10 dB and below, consistent with the interfering words belonging to the target vocabulary. Noise samples differ across SNR levels because the random seed includes SNR; the curves do not isolate changing amplitude on one fixed noise waveform.

5.5 Other languages (exploratory)

The same code runs on English (28 words: 0–19 and the tens) and Modern Standard Arabic (30 words; 25 is *khamsa*

Table 3: Real French accuracy (%) by construction type, Whisper encoder + transformer head.

Type (numbers)	Recordings	Words only	Full numbers
single word (0–16, tens)	475	95.7	95.9
tens + unit (22–69)	277	90.5	94.2
soixante-dix forms (70, 72–79)	43	93.5	95.3
dix-sept/huit/neuf (17–19)	30	80.7	98.9
quatre-vingt forms (80–89)	61	44.6	96.2
et un / et onze (21, . . . , 71)	42	18.6	96.8

Table 4: Real French recordings with added noise; no added noise was used in task-specific training. Real noise: DEMAND [22] kitchen, living room, cafeteria, street. Babble: six TTS recordings sampled from training voices, not necessarily six distinct speakers. Mean of seeds.

SNR (dB)	clean	real 20	real 10	real 0	babble 20	babble 10
mel CRNN-CTC	71.9	22.2	16.9	8.2	17.5	7.7
wav2vec2 + 100-class	92.0	84.8	78.2	56.4	76.0	37.9
Whisper + transformer, full numbers	95.5	93.7	89.0	71.2	73.0	42.7
Whisper + transformer, words only	86.7	85.9	81.6	70.6	71.8	48.8

wa-'ishrūn, "five and-twenty", with *wa*+tens as one word) by changing the word list, the composition rule and the voices (Table 5). English reaches 96.0 % on 249 real recordings from 17 speakers, but its compounds come from 3 speakers. The 8-voice English control gives a similar mean on this collection. This does not isolate a linguistic effect: the languages differ in speakers, number distributions and audio conditions; 125 of the 131 English compound recordings come from two speakers. The Arabic real test contains only 15 compound recordings (11–19), and results vary widely across seeds (27–73 %): it does not support a conclusion.

5.6 Data and computation

The words-only recipe synthesises 736 isolated training clips, versus 3,200 full-number clips, a 4.35-fold reduction in synthesised source files. These counts exclude validation/test clips and the 2,467 accepted augmented sequences. They are not a reduction factor in training examples or compute. The stored transformer head is about 1.9 MB; the shared Whisper-base encoder has 20.6 M parameters (about 82 MB in float32). The archived timing estimates used different accounting boundaries: full-number "training" also included test prediction and checkpoint saving, and extraction estimates counted retained sequences while the implementation encoded rejected ones too. We therefore do not claim measured compute or energy savings. Upstream TTS and encoder pre-training costs are outside this task-level file-count comparison.

6 Discussion

Why are isolated words not enough? Across the synthetic corpus, a word said alone lasts a median 1.55 times longer than the same word in non-final position inside a number (interquartile range 1.38–1.88; 5,668 occurrences, same voice and prosody, from the TTS word timestamps). Time-stretching and cross-fading reduce this mismatch without showing any number; the comparison shows a large differ-

Table 5: Same recipe, Whisper encoder + transformer head, 5 seeds. Training voices: French 8, English 23, Arabic 14. Real test sets are small.

Language	$\|V\|$	Synth. comp.	Real all	Real comp.	Real comp. from
French	23	94.7±0.9	86.7±1.1	77.3±1.8	453 rec., 13 spk
English	28	98.6±1.0	96.0±1.9	94.4±3.4	131 rec., 3 spk
English, 8 voices	28	98.8±0.9	96.0±0.8	95.0±1.6	131 rec., 3 spk
Arabic (MSA)	30	85.7±3.8	69.1±9.6	44±21	15 rec. (11–19)

ence, but does not isolate duration from the other junction cues. The isolated and augmented settings also use different epoch budgets (80 versus 40), so this is not a controlled estimate of the augmentation effect alone. **Where does the gap remain?** The two weak groups are those whose pronunciation inside a number differs most from a concatenation of isolated words: in *vingt-et-un* the final *t* of *vingt* is pronounced and *et un* is reduced, and *quatre-vingt* forms a single prosodic group. This is an interpretation of the error pattern, not a tested cause. **Head and encoder.** BiGRU heads reach high accuracy on held-out synthetic word sequences yet transfer poorly to natural numbers, so this validation criterion does not predict transfer for them; the transformer head transfers much better with both encoders. Why is not established by our experiments.

7 Limitations

Real test sets are small: in French, compounds come from 13 speakers and there is no 90–99; in English from 3 speakers; in Arabic only 11–19. The 95 % speaker intervals are wide (e.g. 71.9–82.1 % on French compounds). Only France numeration is covered. Speech is read and isolated. Test voices of the synthetic test come from the same TTS engine as the training voices. Whisper's pre-training data are not public and could include some Lingua Libre audio. The recipe was developed while monitoring the real evaluation set; it is not an untouched holdout. The ordering of models is not identical on synthetic and real speech. A new, independent speaker cohort is required for confirmatory performance claims. Five French speakers account for 435 of the 928 recordings; the reported aggregate is recording-weighted. Speaker bootstrap intervals describe this collection and do not correct adaptive method selection. The frugality is at the task level: the encoder's pre-training was very costly.

8 Reproducibility

Trained heads, per-run predictions, a fixed French file list and the dependency versions used for the experiments will be released at a later date. All scores here are calculated from the supplied experiment records; the publication cleanup did not retrain the models. An archived seed-0 check reports identical predictions on 928 real and 1,600 synthetic recordings. It is evidence for that recorded environment, not a fresh reproduction of the cleaned code. The audio files, their historical content hashes and the exact original encoder revisions are absent. Regeneration from online TTS/Commons services may differ. The French identity check stops if the fixed file list changes; matching filenames alone cannot establish byte identity with the original corpus. The cleaned code normalises Unicode identities, writes prediction file IDs, binds numeric caches to file-content hashes and processing configurations, and isolates new runs with provenance-checked resumption. Automated unit tests and checks of the archived predictions support the release, but a complete run with reconstructed audio remains necessary to validate end-to-end numerical reproduction.

9 Conclusion

In this study, we showed that a frozen Whisper encoder combined with a CTC transformer head, trained only on synthetic basic words and filtered artificial sequences, achieves 77.3 % accuracy on French compound numbers, compared with 31.3 % when trained on isolated words only. However, these results rely on predefined number construction rules and an encoder already pre-trained on large amounts of speech data. Future work should include evaluation on new, independent recordings to confirm these results and investigate whether this approach can generalise to other languages.

Use of AI tools. AI assistants (Claude, Anthropic; OpenAI Codex) were used to write and correct code, run and check analyses, and draft and revise the manuscript. The author designed the study, made the methodological decisions, checked the results and is responsible for all claims.

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